

Report
Of
**Development of Automated Web Scrapping Systems and Internal
Software Tools**

Submitted
To

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B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, RAJ GUPTA, do hereby declare that the internship report titled “Development of Automated Web Scraping Systems and Internal Software Tools” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I would like to specially thank my mentor for the continuous guidance, patience, and encouragement provided throughout the internship. This support gave me the confidence to independently deploy some of my personal projects even when the assigned organizational projects could not be deployed within the internship tenure.

I also express my sincere gratitude to EduSkills Academy for providing me the opportunity to undertake this internship and for the learning experience it offered.

I express my gratitude to my parents for their blessings.

Signature

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4. Project Description

4.1 Introduction

This report presents an account of the internship undertaken by the author in the domain of Machine Learning & Data Science, completed as a 10-week program conducted by EduSkills Academy. The program covered the Machine Learning and Data Science ecosystem in a structured, week-wise manner — beginning with core Python libraries such as NumPy and Pandas, progressing through data visualisation, supervised and unsupervised learning with Scikit-learn, text data handling and NLP-based text classification, SQL integration, model deployment with Flask, data storytelling and dashboarding with Power BI, and ensemble/hyperparameter-tuning techniques, and concluding with ML pipelines, MLOps concepts, and a capstone project on Customer Sentiment Analysis & Reporting for E-commerce.

Alongside this structured curriculum, the practical component of the internship was delivered through independent, standalone software projects that did not require support from other team members. These projects required end-to-end ownership — development, bug testing, and deployment on servers — and were built using AI-assisted development tools, specifically the Antigravity IDE. The practical assignments centred on building automated data-collection systems (web scrapers) for a Google Maps listing dataset and for a wide range of e-commerce and quick-commerce platforms, which directly fed into the kind of e-commerce data pipeline required for the capstone theme of customer sentiment analysis and reporting. This report documents the curriculum covered, the projects undertaken, the technical approach followed, the challenges encountered, and the outcomes achieved during the internship tenure.

Week	Topics Covered
1	Introduction to Machine Learning & Data Science Ecosystem; Core Python Libraries for Data Science – NumPy
2	Core Python Libraries for Data Science – Pandas; Data Visualisation with Matplotlib & Seaborn
3	Introduction to Scikit-learn & Supervised Learning; Feature Engineering & Preprocessing
4	Model Evaluation & Selection; Unsupervised Learning – Clustering
5	Text Data Handling & Basic NLP; Text Classification with Naive Bayes
6	Introduction to SQL for Data Science; Integrating Python with SQL
7	Model Deployment with Flask; Data Storytelling & Dashboards with Power BI; Advanced Scikit-learn – Ensemble Methods, Hyperparameter Tuning & Model Optimisation

Week	Topics Covered
8–10	ML Pipelines & Best Practices; Introduction to MLOps Concepts; Capstone Project – Customer Sentiment Analysis & Reporting for E-commerce

Table 1.1: Week-wise curriculum of the 10-week Machine Learning & Data Science internship program, as reflected on the internship completion certificate issued by EduSkills Academy.

4.2 Organization Profile

The internship credential was issued by EduSkills Academy, an ISO 9001:2015 and ISO/IEC 27001 certified organisation recognised as a Great Place to Work. EduSkills Academy focuses on bridging the gap between academic learning and industry requirements by offering structured, hands-on training and internship programs in emerging technology domains such as Machine Learning, Data Science, and software automation. The organisation emphasises project-based learning, encouraging interns to independently design, build, and deploy real-world software solutions under the guidance of assigned mentors.

4.3 Problem Statement

Machine Learning and Data Science workflows, such as the customer sentiment analysis and reporting capstone theme covered in the certification program, depend on the availability of clean, structured, real-world data. In the e-commerce and quick-commerce domain, such data — product listings, prices, reviews, and availability — is not readily available in a ready-to-analyse form and must first be collected in a reliable, automated manner from multiple online platforms. Manual collection of this data is time-consuming and error-prone, creating the need for automated web scraping solutions that can feed downstream data science and analytics pipelines. Separately, organisations also require internal tools, such as Human Resource Management Systems (HRMS), that can be customised and self-hosted to suit specific operational needs. The internship was designed to address both needs: building the automated data-collection layer that would support data-science use cases of the kind covered in the curriculum, and developing an internal HRMS tool, while evaluating the practicality of using AI-assisted development tools throughout.

4.4 Project Objectives

- To build a strong foundation in the Machine Learning & Data Science ecosystem, including core Python libraries (NumPy, Pandas), data visualisation (Matplotlib, Seaborn), Scikit-learn based supervised and unsupervised learning, feature engineering, model evaluation, NLP-based text classification, SQL integration, model deployment with Flask, dashboarding with Power BI, and ensemble/hyperparameter-tuning techniques, culminating in a capstone project on customer sentiment analysis and reporting for e-commerce.
- To design and independently develop standalone software projects without requiring support from other team members.
- To build a web scraper capable of extracting business listing data from Google Maps using Python and the Selenium library.
- To design a generic, reusable web scraping framework and extend it to multiple e-commerce and quick-commerce platforms, generating the kind of structured e-commerce data needed for downstream sentiment analysis and reporting.
- To clone an open-source Human Resource Management System (HRMS) from GitHub, customise it, and prepare it for deployment on a personal server.
- To perform bug testing and quality checks on all developed projects prior to deployment.
- To deploy completed projects on assigned servers.
- To explore and utilise AI-assisted development tools, specifically the Antigravity IDE, to improve development speed and code quality.

4.5 Scope of the Project

The scope of the internship spanned three major projects. The first project involved building a Google Maps data scraper. The second, and most extensive, project involved designing a generic web scraping framework and extending it to a wide range of e-commerce and quick-commerce websites, along with an accompanying website to present the scraped data. The third project involved cloning and customising an open-source HRMS for deployment on a personal server. The following table summarises the scope and completion status of the platform-specific scrapers developed as part of the second project:

Website / Platform	Status
Amazon	Completed

Website / Platform	Status
Flipkart	Completed
BigBasket	Completed
Blinkit	Pending (requires location input)
Zepto	Completed
Nykaa	Completed
Myntra	Completed
JioMart	Completed
Purplle	Completed
Tira	Completed
Kult	Pending (website under maintenance)
Smytten	Completed
Alibaba	Not implemented

Table 1.2: Status of platform-specific web scrapers developed as part of the practical, e-commerce data-collection component of the internship.

The Google Maps scraper was limited in scope due to server-side restrictions encountered during data extraction; on reaching these limitations, the project was escalated to the higher management for further evaluation. The accompanying website built to present the multi-platform scraper's data was completed at the front-end level but was not deployed within the internship tenure. The HRMS clone project remained incomplete, as the internship tenure was cut short owing to the resumption of college activities, and hence this project could not be carried through to deployment.

4.6 Technologies and Tools Used

- Python – primary programming language used across the curriculum and for all scraper development.
- NumPy & Pandas – core libraries covered for numerical computing and structured data handling.
- Matplotlib & Seaborn – used for data visualisation as part of the curriculum.
- Scikit-learn – used for supervised and unsupervised learning, feature engineering, model evaluation, ensemble methods, and hyperparameter tuning.
- NLP & Naive Bayes – covered for text data handling and text classification.
- SQL – covered for data storage and querying, and its integration with Python.
- Flask – covered for model deployment.
- Power BI – covered for data storytelling and dashboarding.

- Selenium – browser automation library used to build the web scrapers, including the Google Maps scraper and the multi-platform e-commerce scrapers.
- Antigravity IDE – an AI-assisted development environment used to build the practical software projects, as mandated for the internship.
- GitHub – used to source and clone the open-source HRMS codebase for customisation.
- Web servers – used (or intended to be used) for deployment and hosting of the completed projects.

4.7 System Architecture

The web scraping projects followed a common architectural pattern. A Selenium-driven browser automation layer navigated the target website and rendered its dynamic content. A parsing layer then extracted the relevant structured data, such as product names, prices, availability, and location details, from the rendered page. The extracted data was then organised for downstream use, including presentation through the accompanying front-end website that was developed for the multi-platform scraper project. Each platform-specific scraper was built on top of an initial generic scraping module, which was progressively adapted to handle the structure and anti-scraping measures of individual websites. The HRMS project followed a client-server architecture inherited from the open-source codebase, which was intended to be customised and hosted on a personal server.

4.8 Methodology

- Structured weekly learning: The 10-week program progressed module by module, as summarised in Table 1.1 — from Python fundamentals and core libraries (NumPy, Pandas), through visualisation, supervised and unsupervised learning with Scikit-learn, NLP-based text classification, SQL integration, model deployment and dashboarding, up to ML pipelines, MLOps concepts, and the capstone project on customer sentiment analysis and reporting for e-commerce.
- Independent task assignment: Alongside this curriculum, practical projects were assigned as standalone tasks that did not require collaboration or support from other interns or team members.

- Google Maps scraper: Developed using Python and Selenium; work was carried out until server-side limitations were reached, after which the project was escalated to higher management.
- Generic scraper development: A base scraping module was first built for a generic website structure, which was then extended and customised for each specific e-commerce and quick-commerce platform listed in Section 4.5, generating the e-commerce data foundation relevant to the capstone theme of customer sentiment analysis and reporting.
- AI-assisted development: All practical projects were built using the Antigravity IDE, an AI-powered development tool, as required by the internship guidelines.
- Testing: Each scraper and module was bug-tested individually before being considered complete.
- Mentorship: Regular guidance was received from the assigned mentor, whose support and feedback helped build the confidence needed to independently deploy smaller personal projects.
- HRMS project: The open-source HRMS codebase was cloned from GitHub with the intention of customising and deploying it on a personal server; however, this project could not be completed due to the early conclusion of the internship tenure.

4.9 Expected Outcomes

- A working knowledge of the Machine Learning & Data Science ecosystem, including Python-based data handling, visualisation, supervised and unsupervised learning, NLP-based text classification, SQL integration, model deployment, dashboarding, and MLOps concepts, as certified through the 10-week program.
- A functional Google Maps scraper capable of extracting business listing data, later escalated for review due to server-side limitations.
- A generic, reusable web scraping framework successfully extended to ten out of thirteen targeted e-commerce and quick-commerce platforms, producing structured e-commerce data of the kind used in the certificate's capstone theme of customer sentiment analysis and reporting.

- A front-end website to present the scraped multi-platform data, completed but not yet deployed.
- A customised HRMS ready for personal server deployment (not achieved within the internship tenure due to its early conclusion).
- Practical experience in using AI-assisted development tools (Antigravity IDE) for real-world software development.
- Improved confidence in independent deployment, demonstrated through the successful deployment of smaller personal projects with mentor guidance, even though the primary internship projects could not be deployed within the given timeframe.

4.10 Certificate of Completion

The Internship Completion Certificate issued by EduSkills Academy has been attached in Chapter 3 of this report, in accordance with the required report format.

5. Bibliography/References

Selenium Documentation – Available at: <https://www.selenium.dev/documentation/>

Python Software Foundation – Python 3 Documentation. Available at: <https://docs.python.org/3/>

EduSkills Academy – Internship Credential Certificate, Machine Learning & Data Science, issued 20 March 2026.

Antigravity IDE – AI-assisted development environment used for project development.

GitHub – Open-source HRMS repository used as the base for the internal HRMS customisation project.